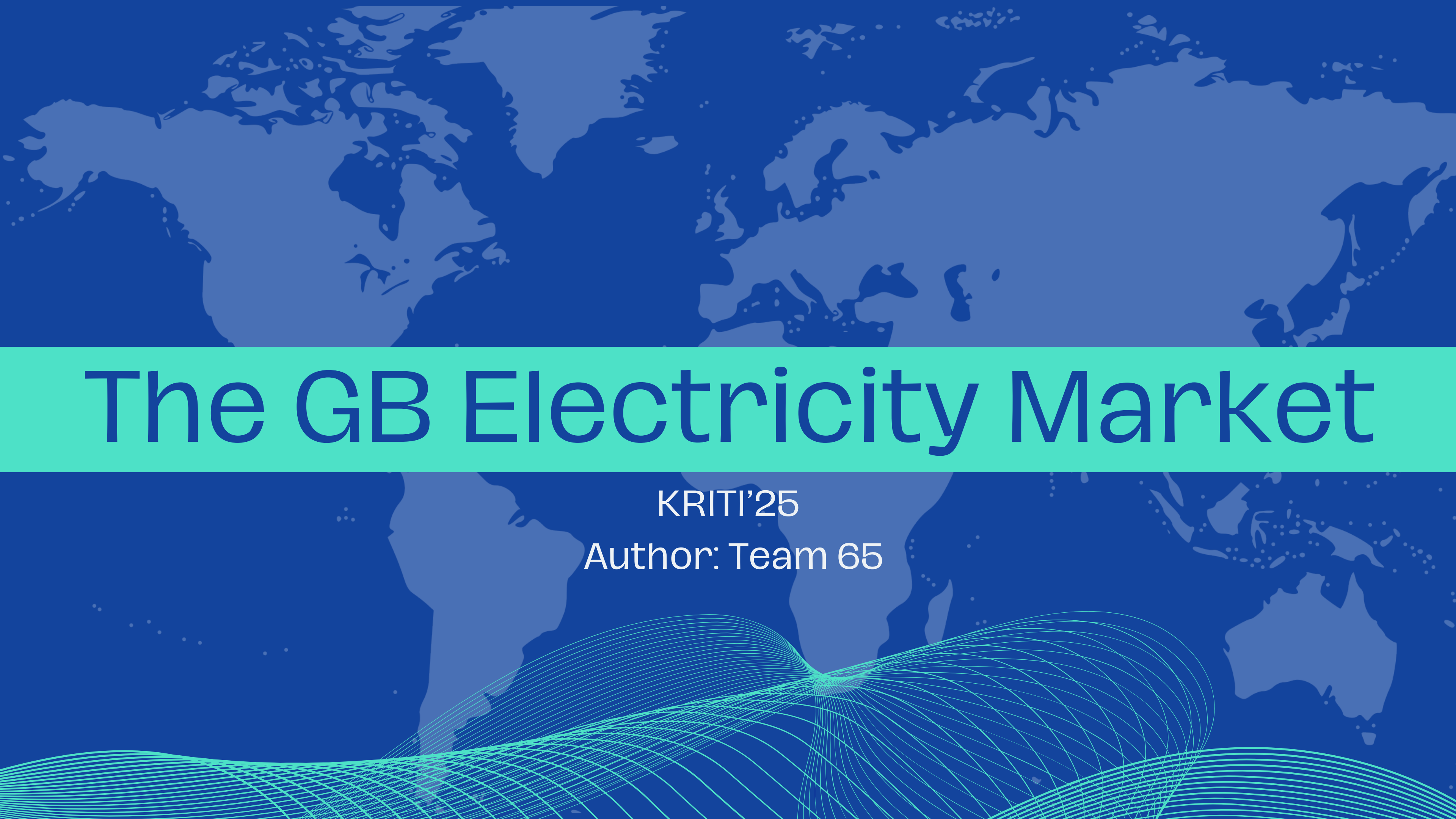




The GB Electricity Market

KRITI'25

Author: Team 65

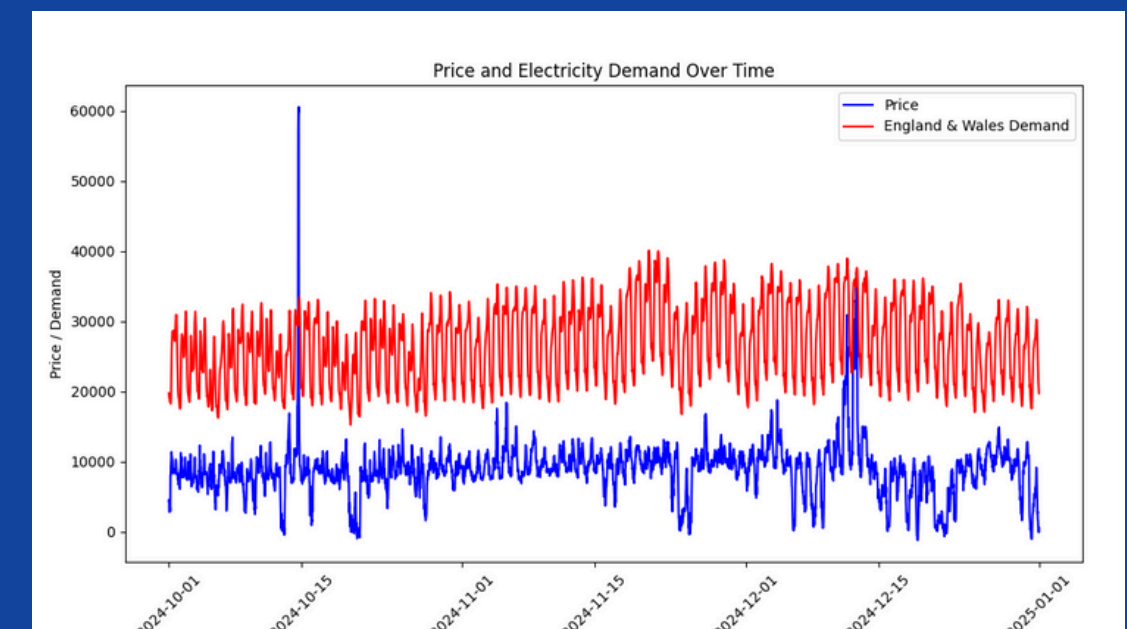
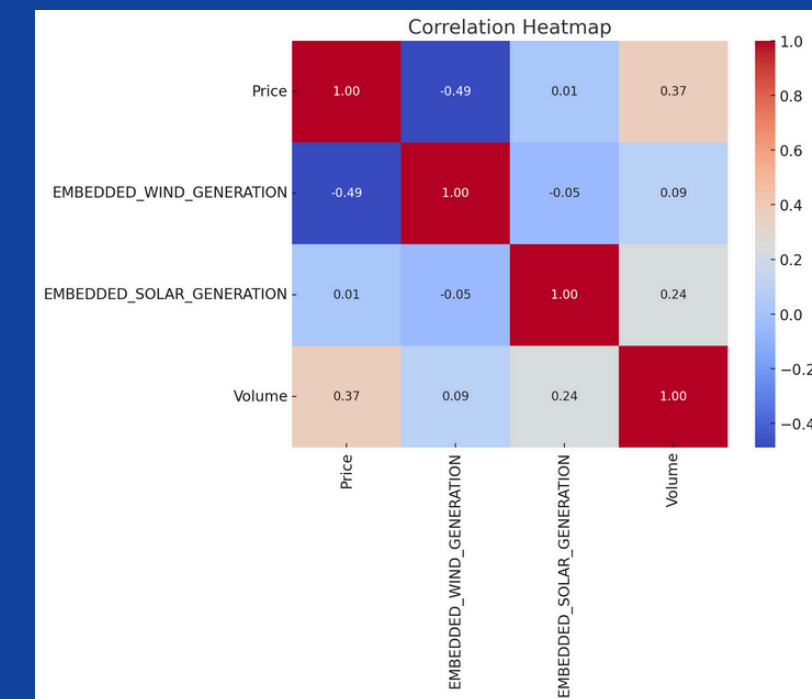
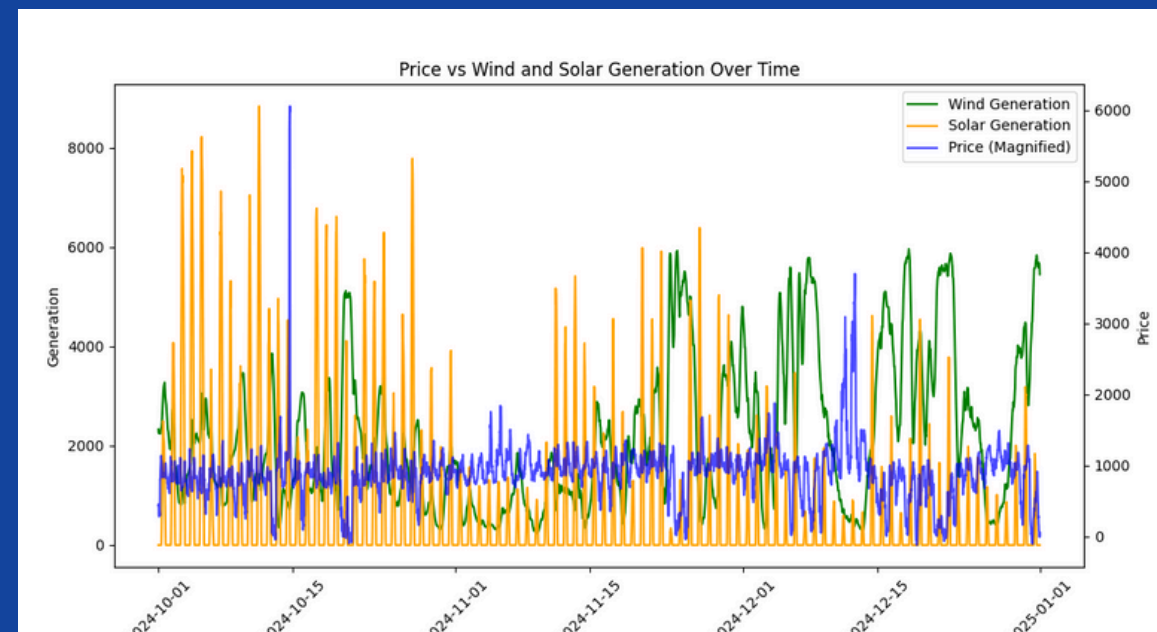
Introduction

The Basic workflow of our model:

- Data Acquisition
- Data Analysis
 - Integration and Analysis of Solar and Wind Data
 - Regime Identification through Hidden Markov Models
- Trading Strategy Development
 - Calculation of Technical Indicators
 - Price Prediction using Long Short-Term Memory (LSTM) Models
 - Implementation of Intraday Trading Strategies using Technical Indicators
- Risk Management Techniques
- Performance Metrics Evaluation

Analysis of Price Trends and Volatility

- **Price vs Renewable Generation:** Prices spike when wind/solar generation is low, revealing dependency on renewables.
- **Rolling Price Volatility:** Highlights high and low volatility periods crucial for trading strategies.
- **Correlation Heatmap:** Shows relationships between variables; price drops with higher renewable generation.
- **Demand-Supply Imbalance:** Imbalances indicate periods of oversupply or shortages, affecting price trends.



Regime Identification through Hidden Markov Models

Regime Classification

We classified the datapoints on the based on the percentage price change from previous day.

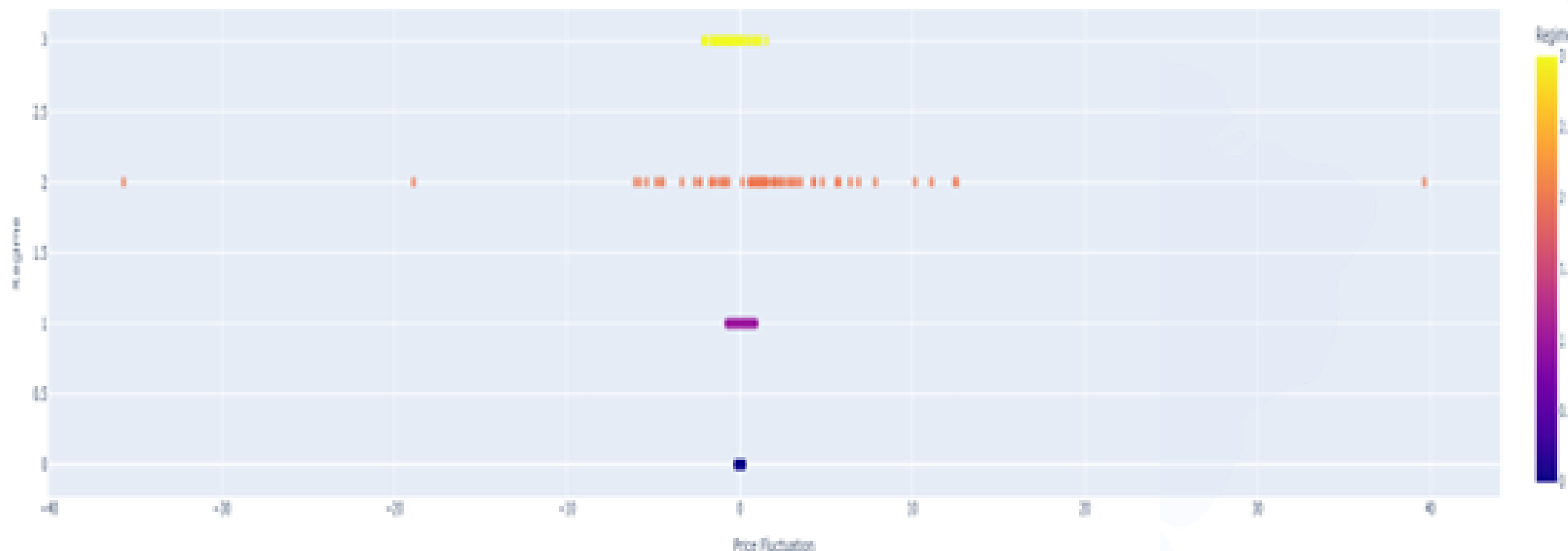
Methods for Regime Classification

we applied K-means Clustering, Hidden Markov Model and Gaussian Mixture Models for Regime Classification.

Conclusion

- K-means failed miserably
- HMM and GMM produced comparable results

HMM MODEL RESULTS

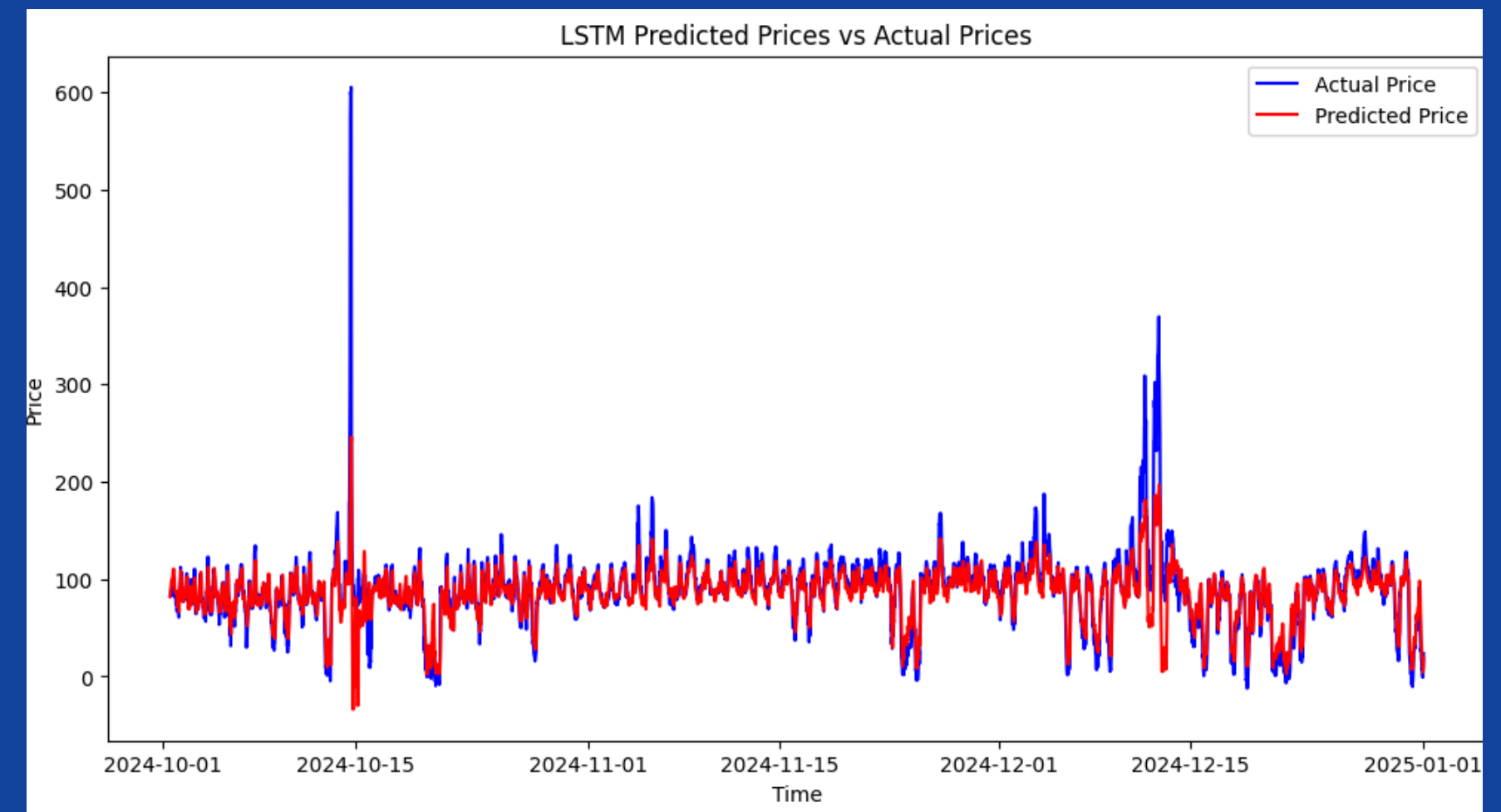


Price Prediction using LSTM

Long Short-Term Memory (LSTM) is used to predict intraday electricity prices based on historical price data and various influencing factors such as wind and solar energy production.

LSTM is a specialized form of Recurrent Neural Network (RNN) designed to capture long-term dependencies in sequential data, making it ideal for time-series forecasting like electricity price prediction.

Other models such as Random forest and ANN were considered but keeping in mind the accuracy and computation cost we decided to go with LSTM



Technical Indicators feeded to LSTM

- We used the technical indicators from the research paper Technical indicators for energy market trading by Catherine McHugh, Sonya Coleman, Dermot Kerr Intelligent Systems Research Centre (ISRC), Ulster University, Northern Ireland, UK.
- The set of 8 Technical indicators used are:
 1. Percentage Price Change Moving Average (PPCMA)
 2. Moving Average Deviation (MAD)
 3. Percentage Range (PR)
 4. Average True Range (ATR)
 5. Relative Strength Index (RSI)
 6. Average Directional Movement Index (ADX)
 7. Moving Average Convergence/Divergence (MACD)
 8. Price Momentum (PMOM)

Trading strategy

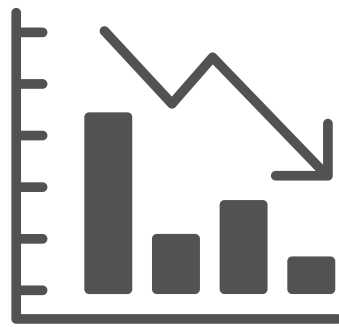
BUY

- Predicted Price exceeds the SMA 24.
- RSI 6 is below the buy threshold (30), signaling oversold conditions.
- MACD is greater than the Signal Line, confirming upward momentum

SELL

- Predicted Price is below the SMA 24.
- RSI 6 is above the sell threshold (70), signaling overbought conditions.
- MACD is below the Signal Line, confirming downward momentum.

Risk Management Techniques



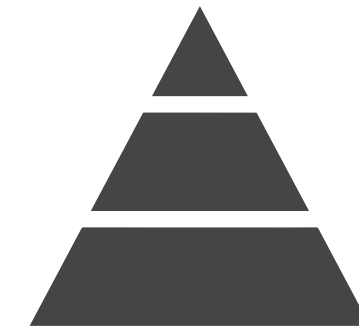
Stop Loss

Stop-loss is set at 1.5 times ATR 24 below (for long) or above (for short) the entry price



Take Profit

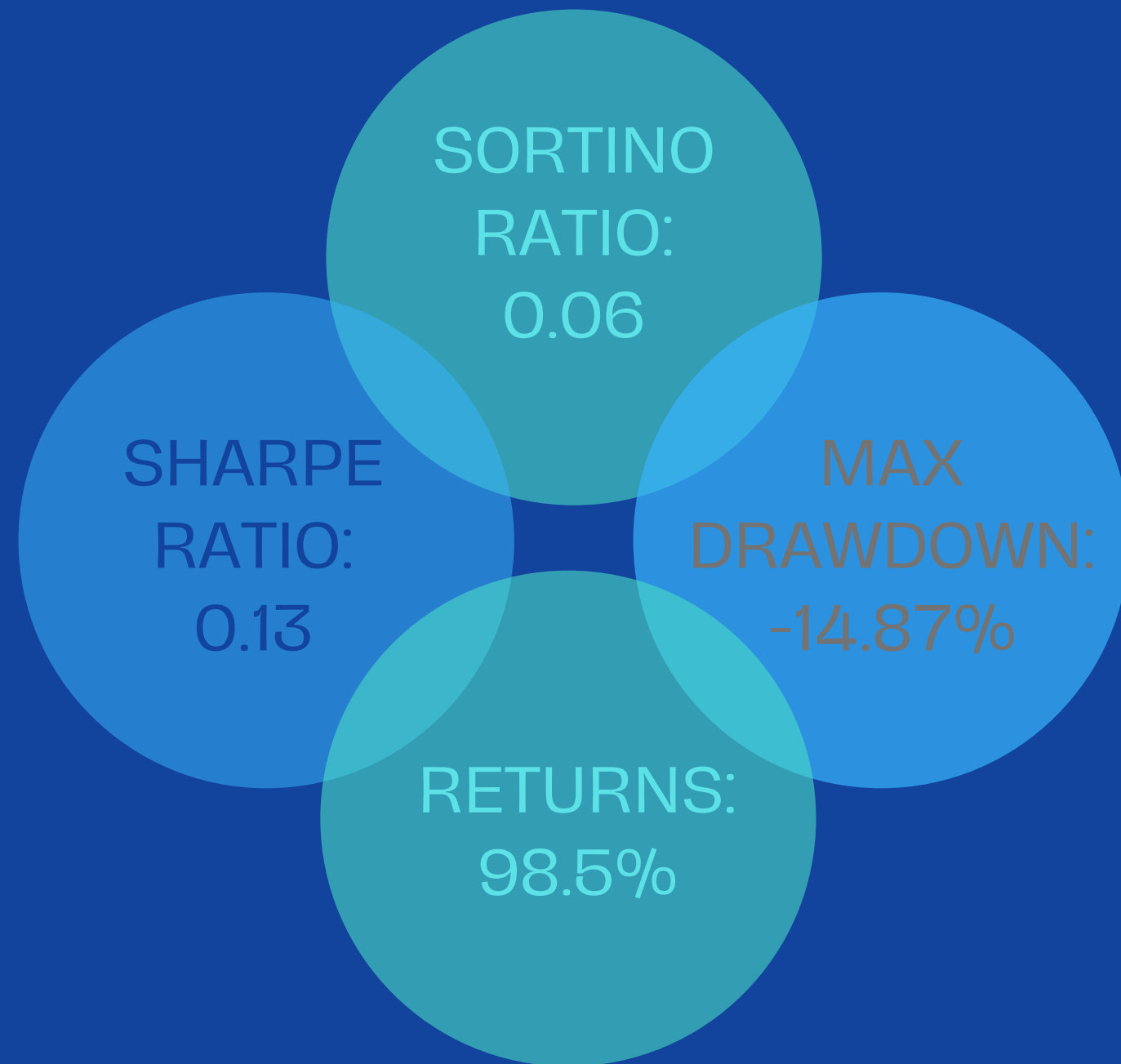
Take-profit is set at 2 times ATR 24 above (for long) or below (for short) the entry price



Position Sizing

2% of capital is risked per trade, with position size adjusted based on the 24-hour ATR to account for volatility

Performance Metrics



The background features a dark blue world map with lighter blue landmasses. At the bottom, there are several sets of thin, light blue lines that curve and flow across the frame, creating a sense of motion or data flow.

Thank You!